LESSON 1: ANCIENT MEDICINE

Mesopotamia, Egypt, Greece, India and China

INTRODUCTION TO MEDICINE | HISTORY OF MEDICINE

What is ancient medicine?

Humanity's earliest efforts to understand illness, relieve suffering and preserve life. It mixed spiritual beliefs, religious rituals, herbal remedies and knowledge gained from observation and experience.

- Five civilizations each built their own explanation of disease: **Mesopotamia, Egypt, India, China and Greece**

ANCIENT MESOPOTAMIA

- **Meaning** – "between rivers": the **Tigris and Euphrates** (modern Iraq, eastern Syria, southeastern Turkey)
- Called the **"cradle of the world's earliest civilization"**

Is Mesopotamia a civilization or an era?

Neither. It is a **geographical region** where several civilizations developed: the **Sumerians, Akkadians, Babylonians and Assyrians**.

WHEN DID IT DEVELOP?

- **Uruk Period** (c. 4000–3100 BCE) – earliest medical practice
- **Old Babylonian Period** (c. 1894–1595 BCE) – an established profession with written texts
- **Cuneiform tablets** – recorded symptoms, treatments and prescriptions

SPIRITUAL VS PRACTICAL MEDICINE

- **Spiritual** – prayers, incantations, spells, amulets, astrology, divination, omens; illness came from **gods, demons or spirits**
- **Practical** – herbal remedies, medicines, wound treatment, surgery, dentistry, cleanliness
- **Remedies** – **willow bark** (and myrtle) as painkiller and anti-inflammatory; honey, wines and salt, animal fat
- No pills; remedies were applied as **poultices**

Note: Mesopotamian medicine was **not purely magic**: rituals and real treatments were used side by side.

PEOPLE INVOLVED

- **Gula** – Sumerian **goddess of healing**; principal deity of medicine (also **Ninkarrak, Ninisinna**); guided doctors and dentists
- **Ninazu** – her son; healing, **serpents**, transformation, underworld; linked to the **rod with serpents**
- **Asu** – **practical physician**: herbs, physical treatment, wound care
- **Asipu** – **spiritual healer**: prayers, incantations, spells, divination; drove away demons like **Lamashtu**

ANCIENT EGYPT

- Developed along the **Nile**, c. **3300–525 BCE**; knowledge kept in **medical papyri**
- Combined clinical treatment with **religion and magic** for illnesses with supernatural causes

KEY FIGURES AND SOURCES

- **Imhotep** – high official under **Pharaoh Djoser**; the most famous Egyptian medical figure
- **Rosetta Stone** – translated in **1822**; let scholars read Egyptian medical records
- **Medical papyri** – Ebers, Edwin Smith, Kahun, Hearst and London

PRACTICES AND TREATMENTS

- **Organized practice** – examine, identify symptoms, diagnose, predict outcome, prescribe
- **Knowledge** – anatomy, diseases, dentistry, wound care, fracture treatment
- **Remedies** – herbs, honey, milk, animal fat, minerals, metals

THE THREE PAPYRI

- **Ebers** – prescriptions, general medicine, diseases, **magical treatments**
- **Edwin Smith** – **injuries, wounds, fractures**, examination, **surgery**
- **Kahun Gynecological** – **women's health**, fertility, pregnancy

ANCIENT GREECE

- Moved medicine toward **rational observation and natural causes**; began clinical documentation
- **5th century BC** – physicians studied symptoms and outcomes while **temple healing continued**

GREEK VIEWS ON HEALTH

- **Asclepius** – legendary healer; sanctuaries like **Epidaurus**; cures through **sacred dreams**; temples called **Asclepieia**
- Divine healing and practical care **coexisted**

FOUR HUMORS THEORY

- **Blood (Sanguine)** – air; cheerful, energetic, optimistic
- **Phlegm (Phlegmatic)** – water; calm, level-headed, deliberate
- **Yellow Bile (Choleric)** – fire; passionate, ambitious, easily angered
- **Black Bile (Melancholic)** – earth; analytical, thoughtful, prone to sadness
- **Cause of illness** – humors become **unbalanced** (poor diet, climate shifts, unhealthy lifestyle)
- **Treatment** – counteract the imbalance and restore **equilibrium**

KEY HISTORICAL FIGURES

- **Hippocrates** (c. 460–370 BCE) – **"Father of Medicine"**; illness has **natural causes**; stressed observation, **prognosis** and **ethics**
- **Galen** (129–c. 216 CE) – Greek physician of the **Roman period**; anatomy and physiology; authority for **over 1,000 years**

MAJOR CONCEPTS AND SOURCES

- **Clinical observation** – monitoring symptoms to guide diagnosis
- **Prognosis** – forecasting the course and outcome of illness
- **Case histories** – documenting patients to build shared knowledge
- **Homer's Iliad** – battlefield wound care in the **Trojan War**
- **Hippocratic Writings** – about **60 treatises**, **not written by Hippocrates** himself
- **Hippocratic Oath** – a doctor's duties

Note: "First, do no harm" is **not** in the original Hippocratic Oath.

ANCIENT INDIA

- **Ayurveda** – "**the science of life**"; over **3,000 years** old; holistic health, herbalism, internal medicine, early surgery
- One of the **oldest continuously practiced** systems; health = balance of the **three doshas**

KEY FIGURES

- **Acharya Charaka** – **Father of Indian Medicine / Ayurveda**; wrote the **Charaka Samhita**
- **Maharshi Sushruta** – **Father of Surgery** and **of Plastic Surgery**; practiced in **Kashi (Varanasi)**
- **Jivaka Komarabhacca** – doctor to **Gautama Buddha**, **King Bimbisara** and the **Sangha**

THE TWO SAMHITAS (SANSKRIT)

- **Charaka Samhita** (4th c. BCE–2nd c. CE) – **internal medicine (Kaya Chikitsa)**; from supernatural to rational diagnosis
- **Sushruta Samhita** (c. 600 BCE–500 CE) – **surgery (Shalya Tantra)** and anatomy; earliest comprehensive surgical treatise

HERBS AND PRACTICES

- **Turmeric** – **curcumin**, anti-inflammatory
- **Neem** – antimicrobial in lab research
- **Garlic** – antimicrobial and cardiovascular effects
- **Holistic diagnosis** – pulse, tongue, eyes, bodily fluids, appearance, mental state
- **Mind-body** – yoga, meditation, chanting
- **Prevention** – daily routine, nutrition, hygiene, exercise, sleep
- **Ethics** – cleanliness, compassion, discipline, dedication

ANCIENT CHINA

- **Traditional Chinese Medicine** – lasted **at least 23 centuries**; restores **yin-yang balance**

HUANGDI NEIJING

- **"The Yellow Emperor's Inner Canon"** – the **foundational TCM text**
- **Suwen** ("Basic Questions") – theory, philosophy, causes, **diagnosis**
- **Lingshu** ("Spiritual Pivot") – anatomy, **acupuncture** channels, needles

CORE CONCEPTS

- **Yin** – passive, feminine, soft, dark
- **Yang** – active, masculine, hard, bright
- Health = **dynamic balance** of yin and yang; **Qi** = energy
- **Wuxing (Five Elements)** – links nature and the body
  - **Creation cycle** – each element nourishes the next
  - **Controlling cycle** – elements control or balance each other

TREATMENTS

- **Herbal medicine** – plant, animal and mineral substances
- **Acupuncture** – thin sterile needles at specific points
- **Mafeisan** – **world's earliest documented herbal anesthetic**; by **Hua Tuo**, 2nd c. AD; taken **orally with wine**; formula **lost** (suspected cannabis, monkshood, datura)

NOTABLE FIGURES

- **Xuanyuan Huangdi** – the **Yellow Emperor** (2697 BCE); qi, yin-yang, wuxing; Daoist deity
- **Shennong** – the **Divine Farmer**; "transparent stomach", **discovery of tea**, "gut-rupturing grass"; **Shennong Bencao Jing**
- **Hua Tuo** (140–208 CE) – physician-surgeon; **Mafeisan**; executed by warlord **Cao Cao**

DIFFERENCES AT A GLANCE

- **Mesopotamia** – gods or spirits; rituals and herbs
- **Egypt** – magic and religion with medicines, wound care, surgery
- **India** – balance of the **three doshas**
- **China** – balance of **yin, yang and qi**
- **Greece** – **natural causes**, observation, **four humors**

Note: Ancient medicine gave us **clinical observation, pharmacology, surgery and medical ethics**.

———— END OF LESSON ————

LESSON 2: THE RENAISSANCE

The Rebirth of Medicine (c. 1400–1700)

INTRODUCTION TO MEDICINE | HISTORY OF MEDICINE

What is the Renaissance?

"**Rebirth**": a general rebirth of culture, art, science and society that began in **Northern Italy**. Its spirit of inquiry, **humanism**, turned medicine toward observation, experiment and the human body.

KEY FIGURES

ANDREAS VESALIUS (1514–1564)

- **Father of Modern Anatomy**; performed **human dissections**
- Corrected **Galen**, who studied **animals**: the **mandible is one bone**; men and women have the **same number of ribs**
- Wrote **De Humani Corporis Fabrica (1543)**; broke Galen's **1,400-year** monopoly

AMBROISE PARÉ (1510–1590)

- **Father of Modern Surgery**; royal surgeon to **four French kings**
- Replaced **boiling oil** with a **soothing balm**; used **ligatures** instead of hot-iron cautery
- Developed **prosthetics** (artificial limbs, mechanical hands and legs)

WILLIAM HARVEY (1578–1657)

- **Father of Physiology**; **circulation of blood (1628)** in a **closed system**
- Disproved Galen's idea that the **liver makes and consumes blood**; used experiments and math
- Wrote **De Motu Cordis**

PARACELSUS (1493–1541)

- **Father of Toxicology**; rejected **Galen and Avicenna's** four humors
- Disease from **external agents**; brought **chemistry** into medicine (mercury, sulfur, iron)
- **"The dose makes the poison"**

LEONARDO DA VINCI

- Artist-anatomist; dissections; drew muscles, bones, organs, **embryology**, nerves

ANATOMY

- Teaching relied on **Galen**, partly based on animals; **human dissection** became central
- Drawings spread knowledge; accurate anatomy was the **first step to surgery and diagnosis**

DE HUMANI CORPORIS FABRICA (1543)

- "**Seven Books on the Fabric of the Human Body**"; based on Vesalius' own dissections; **woodcut** illustrations
- **Books 1–7** – bones and cartilage; ligaments and muscles; veins and arteries; nerves; abdominal and reproductive organs; heart; **brain**

SURGERY

- **Barber-surgeons** – bloodletting with **leeches** and minor surgery
- **Cauterization** – burns tissue with heat; painful and damaging
- **Ligature** – ties the vessel; stops bleeding **without burning**
- **Impact** – observation-based surgery, better wound care, foundation of modern surgery
- **Limitations** – poor sanitation, infection, high mortality

NOTABLE SURGICAL FIRSTS

- **Brassavola (Antonio Branca)** – first **tracheotomy**
- **Zaccarelli** – first **splenectomy**
- **Georg Bartisch** – first **eye removal for cancer**
- **François Rousset** – first **cesarean where the mother survived**

- **Gaspare Tagliacozzi** – **plastic surgery**; nose rebuilt with a **skin graft**

Note: Earlier medicine relied on ancient authority, animal anatomy and tradition. The Renaissance replaced these with **direct observation, human dissection, experiment and detailed drawings**.

PHARMACOLOGY

DRUG THEORIES

- **Reductionism** – drug effects from **hot, cold, wet, dry**
  - **Secondary potencies** – general effects (hardening, softening)
  - **Tertiary potencies** – specific actions (promote urination, dissolve gallstones)
- **Celestial causation** – drug powers come from the **stars**
- **Emergentism** – **mixtures** create effects no single ingredient has

PARACELSUS' CONTRIBUTIONS

- **Arcana** – drugs should **target specific diseases or organs**
- **Three principles of matter** – **sulphur** = combustibility; **mercury** = liquidity/volatility; **salt** = stability
- **Mercury** for syphilis and dropsy; **zinc oxide** ointments
- Laid the foundations of **pharmacology, toxicology and targeted drug therapy**

HOSPITALS, CHALLENGES AND IMPACT

- **Medieval hospitals** – run by the **Church**; **spiritual care over cure**; Galen and superstition
- **Renaissance hospitals** – places of **treatment**; **specialization** (plague hospitals, mental asylums); teaching through **clinical rounds**
- **Challenges** – four humors persisted; resistance to Vesalius; poor sanitation and epidemics; **bloodletting timed by the zodiac**
- **Impact** – better anatomy, knowledge spread by **printing**, observation over **authority-based** knowledge

———— END OF LESSON ————

LESSON 3: 19TH CENTURY MEDICINE

Germ Theory and Public Health

INTRODUCTION TO MEDICINE | HISTORY OF MEDICINE

What changed in the 19th century?

Medicine moved from speculative theories to **laboratory-based science**. It found the **microscopic causes of disease** and recognized that **sanitation and hygiene** prevent it, the start of modern public health.

MIASMA VS GERM THEORY

- **Miasma theory** – disease from **bad air** of rotting matter; wrong, but cleaning and ventilation **worked**
- **Germ theory** – **microorganisms** cause disease
- Miasma was disproved in the **1860s–1880s** by **Snow** (cholera is waterborne) and **Pasteur and Koch** (germ theory)

CHOLERA OUTBREAK

- Originated in India's **Bay of Bengal**; caused by ***Vibrio cholerae***; spread by trade, steamships and armies in **5 waves**
  - **1st (1817–24)** – Asia; 150,000+ dead in India
  - **2nd (1826–37)** – London, New York, Paris
  - **3rd (1846–60)** – **deadliest**; 1M+ in Russia; Snow's **1854** Soho study
  - **4th (1863–75)** – Mecca pilgrims
  - **5th (1881–96)** – **1892 Hamburg**
- **Symptoms** – **rice-water diarrhea**, vomiting up to 1 L/hour; death by **dehydration and hypovolemic shock**
- **Toll** – untreated mortality **50–60%**; Broad Street (1854) **500+ dead in 10 days**; Britain 140,000+ dead in waves 2–3
- **Cholera Riots (1830s)** – doctors accused of poisoning wells to get bodies for dissection

THE GREAT SANITARY AWAKENING (1840s–1880s)

- Care shifted from treating **individuals** to protecting **populations**
- **Filth** seen as both a **cause and a carrier** of disease
- London average age of death: **44** (professionals) vs **22** (working class)
- **Trigger** – the **Industrial Revolution**: migration to crowded cities; smallpox, cholera, typhoid, TB

SANITATION AND DISEASE THEORY

EDWIN CHADWICK (1838)

- **Lawyer** and secretary of the **Poor Law Commission**
- **1842 General Report on the Sanitary Conditions of the Labouring Population** – linked filth, disease and poverty (through miasma)
- **UK Public Health Act 1848** – the **world's first national public health framework**

JOHN SNOW (1854)

- **Father of Modern Epidemiology**; anesthesiologist; **disproved miasma** for cholera
- Cholera struck the **gut, not the lungs**, so it was swallowed
- **Natural experiment (1849–54)** – **Southwark and Vauxhall** (sewage water) vs **Lambeth** (clean upstream): **8x** the deaths, same air
- **Broad Street Pump** – **mapped deaths** in Soho; pump contaminated by a cesspool with an infected baby's diaper
- Largely ignored until **after his death**

GERM THEORY AND MICROBIOLOGY

LOUIS PASTEUR

- French chemist and **pioneer of microbiology**; disproved **spontaneous generation**
- **Swan-neck flask (1861)** – broth stayed sterile until the neck was broken

- Microbes cause **fermentation and spoilage**; **pasteurization** heats wine and milk to kill bacteria
- Disproved **miasma and humoralism**; opened the way to **immunology**

ROBERT KOCH (1882)

- German physician; linked **specific microbes to specific diseases**: **anthrax (1876), TB (1882), cholera (1883)**
- **Agar** media in **Petri dishes** and bacterial **staining**; the "Bacteriological Golden Age"
- **Koch's Postulates (1884)**
  - **1** – present in **every case**
  - **2** – isolated and grown in **pure culture**
  - **3** – culture **causes the disease** in a healthy host
  - **4** – same pathogen **re-isolated**

ANTISEPTICS AND HYGIENE

- **Antiseptics** – kill or stop microbes, e.g. **carbolic acid**
- **Hygiene** – clean habits: washing, sanitation, waste disposal

IGNAZ SEMMELWEIS (1840s)

- Problem: **puerperal (childbed) fever**; a colleague died from an **autopsy** cut
- Doctors carried **"cadaverous particles"**; in **1847** he required handwashing with **chlorinated lime**: deaths fell **18% to 2%**

JOSEPH LISTER (1867)

- **Father of Antiseptic Surgery**; inspired by **Pasteur**
- **Carbolic acid** on wounds and tools; introduced **sterilization**
- Before him: sepsis, gangrene, erysipelas; surgeons wore **blood-stained coats** proudly

FLORENCE NIGHTINGALE (1850s)

- **First female fellow** of the Royal Statistical Society; **Scutari**, **Crimean War**
- **Environmental Theory of Nursing** – clean air, pure water, drainage, cleanliness, light
- **Rose / Coxcomb charts**; **Notes on Hospitals (1859)**

ANESTHESIA

- Before: restraints, alcohol, opium; surgeons judged by **speed**
- **Humphry Davy (1800s)** – English chemist; **nitrous oxide** ("laughing gas") eased **his toothache**; ignored 40+ years
- **Crawford W. Long (1842)** – **first ether anesthesia** (via towel); removed the **neck tumor** of **James M. Venable**; didn't publish
- **Horace Wells (1844)** – dentist; **nitrous oxide**; had his own **wisdom teeth** pulled painlessly
- **William T.G. Morton (1846)** – Boston dentist; **public ether demonstration**

VACCINATION

- **Edward Jenner** – **milkmaids** with cowpox were immune to **smallpox** (Variola, 30% deadly); inoculated a boy; "**vaccination**" from **vacca** (cow); founded **immunology**
- **Louis Pasteur** – first **lab-made, attenuated** vaccines: **chicken cholera 1879** (accident), **anthrax 1881** (sheep), **rabies 1885** (boy bitten by a dog)
- **Emil von Behring (1890)** – **antitoxins / serum therapy** for **diphtheria and tetanus**; **first Nobel Prize in Physiology or Medicine (1901)**

BLOOD TRANSFUSION

- **James Blundell** – **first human-to-human transfusion (1816)**; animal blood had been deadly (banned 100 years); driven by **postpartum hemorrhage**
- Tools: **Impeller** (warmed funnel and syringe) and **Gravitator** (donor to patient)
- Half his patients died, explained later by **Landsteiner's blood types A, B, AB, O (1901)**

DIAGNOSTIC TOOLS

- **René Laennec** – **stethoscope (1816)**: paper tube, then wooden tube, one ear; chest sounds published **1819**; two-ear version 1850s
- **Hermann von Helmholtz** – **ophthalmoscope (1851)**; angled glass plates; first view of the **living retina**
- **Wilhelm Roentgen** – **X-rays (1895)**; X = unknown; first image: **his wife's hand and ring**; **first Nobel Prize in Physics (1901)**

Note: Learn the **event years**: Blundell and Laennec = **1816**, even though their slide headers say 1882.

QUICK REVIEW

FATHERS OF...

- **Medicine** – Hippocrates
- **Indian Medicine / Ayurveda** – Charaka
- **Surgery / Plastic Surgery** – Sushruta
- **Modern Anatomy** – Vesalius
- **Modern Surgery** – Paré
- **Physiology** – Harvey
- **Toxicology** – Paracelsus
- **Modern Epidemiology** – Snow
- **Antiseptic Surgery** – Lister

DON'T MIX UP

- **Asu / Asipu** – practical / spiritual healer
- **Ebers / Edwin Smith / Kahun** – general and magic / injuries and surgery / women's health
- **Charaka / Sushruta** – internal medicine / surgery
- **Suwen / Lingshu** – theory and diagnosis / acupuncture
- **Vesalius / Harvey** – anatomy 1543 / circulation 1628
- **Chadwick / Snow** – 1842 report and 1848 Act / disproved miasma 1854
- **Pasteur / Koch** – microbes in general / one microbe per disease
- **Semmelweis / Lister** – chlorinated lime 1847 / carbolic acid 1867
- **Davy, Wells / Long, Morton** – nitrous oxide / ether
- **Jenner / Pasteur / Behring** – cowpox / weakened lab vaccines / antitoxin serum

————— END OF LESSON —————